

Continue building AMANI.

Developer handoff, completion roadmap and live deployment plan.

Project links

GitHub: [skylarart1720-bot/AMANI](https://github.com/skylarart1720-bot/AMANI)

[Railway project and selected environment](#)

Public website: amani-navy.vercel.app

Moderator website: amani-hck2.vercel.app

The Railway link is a hosting control panel, not an API origin. Copy the public HTTPS API origin from the orchestrator networking settings.

Prepared 3 October 2026

Reviewed against local HEAD feb2462 and the original 14-page blueprint. This document covers implemented features, outstanding work, implementation methods, 13 development prompts, deployment, acceptance and operations.

No production credentials or conversations are included. Railway account settings and current hosted/provider state were unavailable for inspection. Website links come from existing project documentation and remain unverified live.

Sharing

Send this PDF to the developer. Editable HTML and a generation script accompany it. Application code and runtime data were preserved.

Executive assessment and contents.

The project has a substantial application foundation: two Next.js frontends, encrypted support sessions, a human-support queue, editable referrals and knowledge, URL reputation checks and a signed WhatsApp gateway. A complete public service still needs trusted content, individual staff access, live integrations, trained coverage and production verification.

The website uses support.py and support.db. Older main.py ingestion/retrieval and optional PostgreSQL capabilities are not the public support pipeline. DATABASE_URL alone does not migrate live support storage or enable its vector retrieval.

Pages	Subject
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Review evidence

- Local HEAD feb2462: Add Vercel project configuration and deployment guide. Previous commit 0298c57: Initial AMANI application with support and moderator workflows. Remote matches the supplied repo.
- 48 backend tests passed during this handoff; six deprecation warnings; 18.99 seconds. Frontend builds and live integrations were not rerun.
- Existing user-guide assets/scripts were untracked at review time. No applicable AGENTS.md or tracked GitHub Actions workflow was found.
- Railway service settings, volume mounts, domains and deployed SHAs were not inspected. GitHub/public pages could not be fetched through the web reader.

Purpose and intended users.

AMANI is a Ghana-first rights, safety and wellbeing information/referral service. Visitors should reach credible information and routes to accountable human help without registering or supplying identity details.

Pillar	Intended outcome
Protest rights & civic freedom	Rights information and independent legal aid
Activism & youth movements	Civil-society resources and networks
Mental health & wellbeing	Calm support, crisis resources and professional referrals
Climate & environmental justice	Environmental rights and reporting pathways
Digital rights & online safety	Reputation checks and cyber-incident referrals
Youth governance participation	Civic information and participation resources
Human rights defenders	Protection and security support
Gender & intersectional justice	GBV, discrimination and witchcraft-accusation support
Men's Circle	Non-judgemental counselling and peer support

Service boundaries

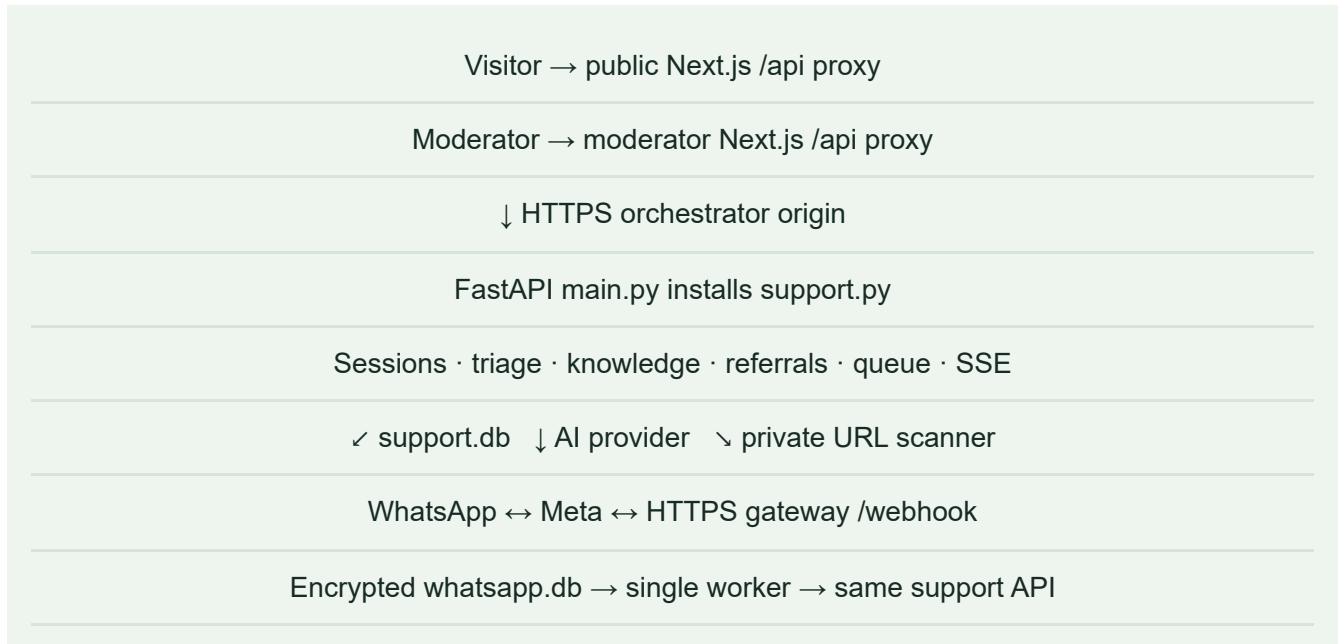
AMANI does not guarantee emergency intervention, legal representation, treatment, partnership acceptance or immediate human response. A directory and queue do not establish a staffed service.

Completion has two layers

Launch readiness means all advertised channels are dependable, private, reviewed and staffed. Full blueprint delivery additionally includes Ghanaian languages, scheduled reviewed knowledge, aggregate reports, richer referrals, consented follow-up and later SMS/USSD. Agree these milestones separately.

Witchcraft-accusation work crosses multiple pillars: approved content and actual referrals are needed, not only a topic description.

Architecture and hosting model.



Single-node baseline

Use one orchestrator instance, one Uvicorn worker and one mounted SQLite volume. Scanner stays internal. Gateway needs persistent storage and one worker. Vercel hosts frontends; its server proxy needs an externally reachable HTTPS API because it cannot access Railway private hostnames.

Realtime and scale

Visitor/moderator SSE checks changes once per second. Clients reconnect and poll as fallback. Validate hosted buffering and duration limits. Do not enable replicas or multiple gateway workers until storage, job coordination and limits are shared.

Privacy

Fernet encrypts stored message content; visitor capability tokens are hashed. The server and authorised staff can read chats. Web chat is not end-to-end encrypted against AMANI. Staff currently share a bearer token via a secure HTTP-only dashboard cookie.

Repository map.

Path	Responsibility
<code>apps/web/app/page.tsx</code>	Chat, handoff, topics, referrals, scanner and privacy
<code>apps/web/app/localization.tsx</code>	English/French strings
<code>apps/web/app/api/[...path]/route.ts</code>	Allowlisted public proxy
<code>apps/moderator-dashboard/app/page.tsx</code>	Login, queue, replies, directory/knowledge editing
<code>apps/moderator-dashboard/app/api/[...path]/route.ts</code>	Cookie auth and admin proxy
<code>apps/orchestrator/src/support.py</code>	Active encrypted support/admin workflow
<code>apps/orchestrator/src/main.py</code>	FastAPI app and legacy knowledge/data/PostgreSQL
<code>apps/orchestrator/src/approved_source_connector.py</code>	Legacy source collection; not scheduled public-content ingestion
<code>services/url-safety/src/main.py</code>	Reputation lookup and scans
<code>apps/whatsapp-gateway/src/main.py</code>	Signed webhook, encrypted inbox, retry worker
<code>packages/shared/src/index.ts</code>	Shared types; check against active API
<code>docker-compose.yml</code> / <code>Dockerfiles</code>	Container wiring and volumes
<code>scripts/</code> and <code>service tests/</code>	Launcher, browser/integration/regression checks

Read order

README.md → PRODUCTION.md → VERCEL.md → support.py → frontend proxies/pages → gateway/scanner → tests → blueprint_text.txt. Preserve /support and /admin contracts; legacy /chat is not interchangeable.

Implemented features and evidence.

Feature	Source-backed behaviour
Visitor UI	Responsive nine-topic interface; English/French; no registration
Sessions/privacy	Random capability, hashed token, encrypted messages, seven-day expiry, delete and quick exit
AI	Explicit consent, recent context, provider integration and labelled directory fallback
Content	Persisted editable referrals; pending/published/rejected knowledge selected by topic
Urgent routing	Bilingual rules before AI; crisis resource and critical case
Human help	Queue/transcript/replies; queued/in_progress/resolved
Realtime	Visitor/admin SSE with reconnect/poll fallback
Staff access	Shared token; HTTP-only same-site one-hour cookie
URL reputation	Safe Browsing/VirusTotal; consent; fresh/pending scan; unknown on failure
WhatsApp intake	Verify token, raw-body HMAC, destination validation, encrypted inbox and duplicate suppression
WhatsApp commands	human, updates, enable ai, disable ai, forget; text/retry delivery

Verification

Current rerun: 48 backend tests passed. Existing VERIFICATION.md records earlier builds, browser checks at eight widths, local SSE, origin/login regressions and VirusTotal results. Historical records are not hosted verification.

Implemented, configured and verified live are distinct. Key presence does not prove credit; a queue does not prove staffing; a seed contact does not prove partnership. Quick-exit deletion can fail during connectivity loss.

Remaining work and priorities.

Gap	Recommended completion
P0 · Staff identity	Individual accounts, MFA, roles, revocation and actor/resource audits before multi-staff real data
P0 · Trusted content	Review all nine pillars, emergency contacts, translations and availability
P0 · Production operations	HTTPS/persistent disks, funded providers, trained coverage, alerts, backup/restore proof
P1 · AI grounding/triage	Relevant retrieval, provenance, structured validation and expert safety evaluation
P1 · Knowledge pipeline	Unify public/legacy stores, scheduled allowlisted ingestion, revisions, review and freshness
P1 · WhatsApp	Proactive human replies, receipts, consented templates, topic menu and language
P1 · Referrals	Optional region, coverage/language, trust tiers, evidence/review dates
P1 · Languages/reports	Twi/Ga/Ewe/Hausa with professional review; aggregate outcomes and feedback
P2 · Scaling	Active support PostgreSQL migration, shared broker/outbox, distributed limits
P2 · Later scope	SMS/USSD; phone/screenshot-text scams; local registry; consented incident reporting

Source does not establish formal NGO partnerships, independent security/privacy review, certified accessibility, source freshness SLA or web E2EE. No proactive WhatsApp human delivery, Ghanaian languages, aggregate dashboard or consented case-worker contact workflow is demonstrated.

Agree public pilot scope with the owner. Do not advertise later blueprint features before delivery. Professional/legal/clinical sign-off is a delivery task, not an automated feature.

Milestones and work packages.

Milestone	Exit evidence
M0 · Reproduce	Fresh installs, both builds, current tests, synthetic local handoff, recorded SHA/dependencies
M1 · Live staging	Railway services/volumes, two Vercel projects, provider checks, restart persistence
M2 · Trust/operations	Staff identity/audits, reviewed content, verified referrals, trained escalation drill
M3 · Complete channels	Grounded knowledge, reliable WhatsApp human replies, regional matching, hosted outage/reconnect tests
M4 · Inclusion/reporting	Reviewed Ghanaian languages, analytics/feedback, accessibility/low-bandwidth evaluation
M5 · Scale/expand	Shared PostgreSQL/jobs/limits, load evidence, SMS/USSD and partners

First assignments

1. Developer: reproduce baseline and inventory actual Railway/Vercel settings without touching real data.
2. Developer + operations: staging, identity, alerts, WhatsApp outbound and restore tooling.
3. Editorial/legal/clinical reviewers: approved content, emergency resources, referral verification and safeguarding.
4. Developer + reviewers: ingestion/retrieval, language coverage and evaluations.
5. Owner: pilot hours, incident ownership, scope exclusions and service measures.

Estimate after access/content availability is known. Meta onboarding and professional reviews can dominate delivery. Do not inherit original blueprint week ranges as promises.

Use small reviewed PRs, protected branches and staging. Each package includes tests, acceptance evidence and migration/rollback where applicable.

Local developer setup.

Fresh Windows checkout

```
git clone https://github.com/skylarart1720-bot/AMANI.git
cd AMANI
py -3.12 -m venv .venv
.\.venv\Scripts\python.exe -m pip install -r apps/orchestrator/requirements.txt -r services/url-
safety/requirements.txt -r apps/whatsapp-gateway/requirements.txt
.\.venv\Scripts\python.exe -m pip install -r scripts/test_requirements.txt
npm.cmd --prefix apps/web ci
npm.cmd --prefix apps/moderator-dashboard ci
Copy-Item .env.example .env
npm.cmd run build
.\.venv\Scripts\python.exe scripts\run_local.py
```

Copy .env only in a fresh checkout and populate privately. Use a supported Node release compatible with locked Next.js and pin it in CI. Containers use Python 3.12; earlier Windows verification used 3.11. Launcher prints actual free ports.

Default address	Purpose
127.0.0.1:3000	Web
127.0.0.1:3100	Moderator
127.0.0.1:8100	API
127.0.0.1:8200	Scanner
127.0.0.1:8300/webhook	Local WhatsApp; not accessible to Meta

Login uses ADMIN_API_TOKEN or generated data/.admin-token; key in data/.chat-key. Do not transfer these. Process records/logs: .runtime/. Stop with scripts/stop-local.ps1.

Launcher is Windows-oriented. On Linux/macOS use python3/.venv/bin/python for tests; Compose is the portable stack starting point.

API and proxy contract.

Browsers call same-origin /api/...; Next.js strips /api before forwarding. Visitor routes use the random bearer capability. Admin routes require staff bearer auth.

FastAPI route	Method / function
/health	GET · process health, not provider readiness
/support/status	GET · configured flags and limits
/support/directory	GET · topics/referrals
/support/sessions	POST create; DELETE with visitor token
/support/messages /support/events	GET transcript/case or SSE; visitor token
/support/chat	POST message, topic?, ai_consent, language en/fr
/support/handoff	POST human-help request; visitor token
/support/check-link	POST URL request with consent
/admin/queue /admin/events	GET staff queue/SSE
/admin/queue/{case_id}	PATCH queued/in_progress/resolved
/admin/queue/{case_id}/reply	POST human reply
/admin/directory	GET/PUT list/upsert
/admin/knowledge	GET/POST list/pending entry
/admin/knowledge/{id}	PATCH published/rejected

Dashboard auth

POST /api/login trims token whitespace, checks /admin/queue and sets amani-admin; /api/logout clears it. Proxy forwards cookie token server-side. Vercel does not need ADMIN_API_TOKEN.

Chat/reply maximum: 4,000 characters. Preserve modes directory, ai, urgent and human. Origin check compares browser Origin with Host: diagnose proxy/domain mismatch rather than disabling it.

Shared types in packages/shared need alignment as fields evolve. Keep backward-compatible response shapes for both web and gateway.

Storage, retention and migration.

Store	Meaning
support.db	Active sessions/messages/cases/audit/directory/knowledge
whatsapp.db	Encrypted inbox/session token, HMAC sender lookup, retries
amani_line.db / legacy PostgreSQL	Older chats/knowledge/vector capabilities; legacy chat may be plaintext

Lifecycle

Sessions expire seven days from creation; hourly/request-time cleanup. Messages/cases cascade on delete; audit retention up to 90 days. Gateway inbox/sender retention seven days; sent payloads cleared. Backups/provider/device records need separate policy.

Secrets

Keep `CHAT_ENCRYPTION_KEY` stable and recoverable separately. Replacement without decrypt/re-encrypt migration breaks ciphertext. Add key IDs and tested rotation before changing it. Staff token is read at import: restart/redeploy for rotation.

PostgreSQL upgrade

1. Introduce support-store abstraction behind active routes, preserving tokens/encryption/deletion/API semantics.
2. Create versioned schemas/indexes for support/gateway/outbox. `DATABASE_URL` alone is insufficient.
3. Migrate authorised snapshot; verify counts, transcripts, ciphertext and case links.
4. Pause writes for cutover; retain compatible rollback/key; test expiry/delete/reconnect.
5. Add shared jobs/limits and concurrency evidence before multiple replicas/workers.

Use fresh staging data. Do not upload local data/ wholesale. AMANI deletion cannot erase Meta/device or unmanaged backup copies; document these limits accurately.

Environment configuration.

Variable	Service / setting
APP_ENV	API/scanner: production
PROJECT_ROOT	Python containers: /app
AMANI_DATA_DIR	API/gateway: /data with persistent mount; never blank
CHAT_ENCRYPTION_KEY	API/gateway: stable Fernet secret
ADMIN_API_TOKEN	API only: fresh high-entropy staff token
URL_SAFETY_URL	API: actual private scanner origin incl. 8001
OPENAI_API_KEY / MODEL	API: funded key and supported model; verify source default gpt-4o-mini
OPENAI_BASE_URL	API: https://api.openai.com/v1 unless approved override
ORCHESTRATOR_URL	Web/admin: public HTTPS API; gateway: private API
COOKIE_SECURE	Moderator app: true
SAFE_BROWSING_API_KEY / VIRUSTOTAL_API_KEY	Scanner: permitted-use credentials/quota
WHATSAPP_PROVIDER	Gateway: meta only after onboarding
META_WHATSAPP_TOKEN / PHONE_NUMBER_ID	Gateway: approved access credential and provisioned number ID
META_WEBHOOK_VERIFY_TOKEN / APP_SECRET	Gateway: challenge token and raw-body HMAC secret
META_WHATSAPP_API_VERSION	Gateway: supported version from Meta app dashboard

Generate secrets privately

```
.\.venv\Scripts\python.exe -c "import secrets; print(secrets.token_urlsafe(32))"  
.\.venv\Scripts\python.exe -c "from cryptography.fernet import Fernet;  
print(Fernet.generate_key().decode())"
```

First output: staff token. Second: Fernet key. Transfer through secret settings, never documents/tickets/repo. Keep staging/production separate.

No NEXT_PUBLIC_* secrets/API origin. Gateway default API port is 8005; Docker API uses 8000. Always explicitly configure ORCHESTRATOR_URL.

Railway deployment runbook.

Open [the provided environment](#). Inventory existing services, source SHA/branch, root, health, domain and mount first; reuse correct services.

Service	Root / Dockerfile / fixed port
orchestrator	apps/orchestrator · Dockerfile · 8000
url-safety	services/url-safety · Dockerfile · 8001
whatsapp-gateway	apps/whatsapp-gateway · Dockerfile · 3001

1. Connect repo/branch. Set each Root Directory to the service directory: COPY paths assume this context. Confirm Docker detection.
2. Scanner: provider settings + APP_ENV=production; /health; private only.
3. API: persistent /data; page 12 variables; actual scanner private origin incl. 8001; one replica/Uvicorn worker.
4. Health /health; align Railway PORT/target with Docker 8000. Generate public HTTPS API domain for Vercel server proxy.
5. Gateway: separate persistent /data volume; Meta settings/key; private API origin incl. 8000; one worker/replica; HTTPS target 3001.
6. Deploy scanner→API→gateway; synthetic tests prove internal connectivity and external HTTPS.
7. Restart/redeploy API/gateway and prove persistence; record real API/webhook URLs.

Railway does not automatically share Compose support-data. Gateway needs whatsapp.db on its own disk and explicit key; it calls support API, not support.db.

Use assigned private hostnames. If IPv6 DNS cannot reach supplied 0.0.0.0 bind, adjust IPv6-capable binding according to platform guidance and test. Keep scanner/disks private.

Official references: [Dockerfiles](#) · [Volumes](#) · [Private networking](#) · [Healthchecks](#).

Vercel deployment runbook.

Import repository twice or inspect existing projects. Existing guide URLs: amani-navy.vercel.app and amani-hck2.vercel.app; confirm ownership/SHA and live behaviour.

Setting	Public	Moderator
Root	apps/web	apps/moderator-dashboard
Framework	Next.js	Next.js
Install / build	npm ci / npm run build	npm ci / npm run build
Output	Framework default	Framework default
ORCHESTRATOR_URL	Railway public HTTPS API	Same API
COOKIE_SECURE	Not needed	true

1. Set Preview/Production independently. Localhost and Railway dashboard link are invalid API origins.
2. Deploy web; /api/support/status must return backend JSON; send fictional chat.
3. Deploy moderator; enter current backend token and verify queue/reply. No ADMIN_API_TOKEN variable needed in Vercel.
4. Verify same-origin mutations, secure cookie, invalid token and logout; check Host/Origin on custom domains.
5. Interrupt SSE and test host duration limits; polling/reconnect must deliver human replies.
6. Configure owned domains/HTTPS; restrict moderator/demo access and verify protection on exact shared URL.
7. Redeploy after settings changes; record URLs and frontend/backend SHAs.

Alternative host

docker compose up --build -d supplies portable baseline. Use HTTPS reverse proxy before loopback web/admin/webhook ports; keep API/scanner internal, persist support-data, validate containers/streaming/backups. Vercel does not run Compose.

References: [Vercel monorepos](#) · [Deployment protection](#).

Activate AI and URL reputation.

AI

1. Assign budget/account owner, fund provider and configure key/model/base URL on API.
2. Restart/redeploy; run scripts/check_integrations.py. ai_configured only means key presence.
3. Fictional hosted consent-on message must return mode=ai; consent off must make zero model calls.
4. Test quota/key/timeout fallback and urgent bypass; set budget/error alerts without transcript logging.

Historical verification reports 429 credit_balance_exhausted. This is not a current balance check; retest actual provider.

Scanner

1. Configure approved provider(s), private scanner URL and /health connectivity.
2. Test consent, current/missing/stale report, fresh/pending/completed scan and provider failures using mocks/sanctioned URLs.
3. Preserve unknown on missing evidence; no_known_threats is not guaranteed safety.
4. Explain VirusTotal exposure of private URL tokens; pending jobs in process memory can be lost on restart.

Scanner never visits submitted URL. Private/local inputs are rejected. Phone/screenshot-text scams, local campaigns and consented CSA reporting remain separate work.

References: [Safe Browsing](#) · [VirusTotal scan](#) · [URL reports](#). Check actual quotas/terms.

Activate WhatsApp and human handoff.

1. Complete Meta business/app/number onboarding, production permissions, token owner/rotation and supported Graph version.
2. Gateway persistent disk/one worker; /health configured=true; private API reachable.
3. Register <https://ACTUAL-GATEWAY-DOMAIN/webhook> and matching verify token; subscribe to messages.
4. Correct GET challenge succeeds; invalid HMAC/wrong destination POST fails.
5. Authorised test phone: fictional text, welcome, consent off; test enable ai/disable ai/human/updates/forget.
6. Replay inbound ID; mock delivery interruption/retry/terminal failure and confirm operator visibility.
7. Before complete WhatsApp launch add proactive moderator outbox, status receipts, permitted templates, menu and languages.

Current limitations

Human replies are pulled with updates. No proactive template flow or interactive nine-topic menu. Gateway defaults to English. Delivery is at least once; a timeout after accepted send may duplicate a retry. Saved answer reduces repeated processing but support effects need stronger idempotency.

Provider policy and privacy

Verify current customer-service window, template categories, fees and opt-in rules directly with Meta. Do not hardcode remembered policy. Meta sees phone number; AMANI encrypts payloads and hashes sender lookups. forget cannot delete Meta/device messages.

Test expired sessions consistently for commands and chat. Failed jobs, invalid tokens and callback status need operational alert/retry controls.

[Meta Cloud API overview](#) is the starting reference; fetch failed during review, so production account details require verification.

Acceptance and launch gates.

Repeatable checks

```
npm.cmd run build
.\.venv\Scripts\python.exe -m pytest apps/orchestrator/tests services/url-safety/tests apps/whatsapp-gateway/tests -q
.\.venv\Scripts\python.exe scripts\check_integrations.py
.\.venv\Scripts\python.exe scripts\browser_check.py
.\.venv\Scripts\python.exe scripts\origin_check.py
.\.venv\Scripts\python.exe scripts\realtime_check.py
.\.venv\Scripts\python.exe scripts\check_moderator_login.py
```

Current rerun: 48 passed, six warnings, 18.99s. Builds/provider/browser checks were not rerun. Local scripts require running services; parameterise staging before hosted acceptance. Never use production fixtures.

Hosted acceptance

- Visitor isolation; expiry/delete cascades and active stream updates.
- Consent-off zero model calls; funded AI/fallback and urgent bypass.
- Request/queue/staff reply/visitor response/resolution; auth/logout/revocation.
- Pending/withdrawn content excluded; honest referral verification.
- English/French/mobile/keyboard/screen reader/low bandwidth/quick exit.
- URL consent/pending/unknown; WhatsApp signature/dedup/retry/forget.
- Persistence after restart; backup restore; timeout/reconnect; abuse/outage.

Block launch for unverified emergency contacts, advertised unstaffed support, lost keys/data, unauthorised access, failing deletion or fabricated safety. Obtain content/safeguarding/privacy/security review.

Operations and rollback.

Owners and staffing

Name deployment, incident, privacy, content and safeguarding owners. Publish true hours/response limits. Confirm referral numbers/hours/languages directly; agree escalation contacts. Train staff and run synthetic urgent drills.

Monitoring

Alert on uptime, API/provider errors, queue/critical age, failed WhatsApp jobs, disk and spend. Redact content/phone numbers. Safe retry/acknowledgement and editorial freshness tasks need operator ownership.

Backup and recovery

Use consistent SQLite backup, not live file copies. Protect support.db/whatsapp.db and keys separately. Set retention/access/recovery targets; isolated restore must prove ciphertext, cases and gateway state. Retain deletion obligations when restoring old data.

Release sequence

1. Record tested SHA/schema versions; recoverable pre-migration snapshot.
2. Compatible API first, frontends next; synthetic smoke immediately.
3. On failure restrict affected flow and redeploy last compatible version.
4. Do not run old code over incompatible schema. Reviewed recovery and deletion tombstones prevent resurrecting erased chats.

Cost and capacity

Budget hosting, persistent services/disks/backups, AI tokens, WhatsApp fees, monitoring and staffing. Obtain account prices and measure usage; no total is implied without traffic/staff assumptions.

Content refresh, updates, staff supervision and drills continue after launch.

Troubleshooting and review risks.

Symptom	Action
API 503	Check actual origin, backend, DNS/HTTPS/timeouts; redacted logs
Startup fails	Validate production token/Fernet key and writable /data
Data lost	Verify persistent volume and AMANI_DATA_DIR
Shared 429	Trusted proxy attribution and shared rate limits
Login failure	Current token, API, cookie; do not print secrets
Mutation 403	Host/Origin domain mismatch; preserve checks
SSE stalls	Buffering/time limits; prove reconnect/polling
AI unavailable	Key/model/quota/provider; safe health categories
Scanner unknown	Missing evidence/provider failure/lost job; preserve uncertainty
WhatsApp silent/duplicate	API/settings/jobs/policy; outbox/receipt reconciliation

Hardening targets

API reads body before size check: add bounded receive and edge limits. Audit stream/transcript access with actor/resource/outcome. Test ambiguous topics; grounding/output validation must extend beyond prompts.

No tracked CI found. Verify next lint tooling. Historical backend dependency audit incomplete. Six current test warnings concern lifecycle/dependency deprecations; plan deliberate maintenance.

Next.js origin checks do not alone secure the externally reachable API. Review edge abuse/session/provider quota protection.

Prompts: identity and audits.

Prefix any brief on pages 20–25 with this shared block.

Work in AMANI. Read README.md, PRODUCTION.md, VERCEL.md, blueprint, support.py and proxies. Extend active /support and /admin, not silently legacy /chat. Preserve anonymity, consent, deletion and conservative failures. Use synthetic fixtures and redact secrets/content. Deliver code, migrations, tests, updated config/docs and acceptance evidence.

A · Individual staff identity

Copyable brief: Replace shared token with approved OIDC accounts, moderator/editor/reviewer/admin roles, MFA, expiry/logout and immediate revocation. Enforce API/proxy/SSE permissions. Controlled time-limited migration; secrets server-side.

Acceptance: Wrong-role access fails; revoked staff lose REST/stream access; reviewer-only publication; hosted login/logout passes.

Start: support.py; moderator page/proxy.

B · Attributable audits

Copyable brief: Record actor/action/object/time/outcome for transcript/stream reads, replies, case/referral edits and reviews. Exclude raw text/tokens/phones. Authorised search, integrity and retention.

Acceptance: Distinct staff attributable; failed access safely recorded; retention works; staff cannot edit audits.

Start: support.py audit/admin; dashboard.

Prompts: knowledge and safe AI.

C · Reviewed-content pipeline

Copyable brief: Unify legacy/public knowledge. Scheduled allowlisted ingestion, revision hashes, metadata, reviewer identity, publication/withdrawal/freshness. Bound fetches and prevent private/local/redirect SSRF. New revisions remain pending.

Acceptance: Pending/withdrawn excluded; idempotent ingestion; provenance/reviewer and revisions; stale flags/retries.

Start: main.py; approved_source_connector.py; support.py; knowledge UI.

D · Grounded retrieval and triage

Copyable brief: Replace three topic-only entries with relevant published retrieval using language/optional region. Version prompt; structured claims/cited entry/referral IDs/outcome; validate provenance and support. Preserve urgent rules before AI; expert bilingual/adversarial evaluation.

Acceptance: No invented facts/contacts/citations; no consent means no calls; withdrawal immediate; urgent bypass; missing evidence fallback; expert false-negative review.

Start: support retrieval/ai_reply/chat; tests; visitor UI.

Vector search does not itself prove accuracy; reviewed evidence and measured grounding remain required.

Prompts: WhatsApp completion.

E · Proactive human delivery

Copyable brief: Durable outbox on WhatsApp case replies; encrypted destination mapping, opt-in and last inbound time. Permitted session/approved template sending under current Meta rules. Provider IDs/status callbacks/backoff/failed jobs/operator retry. Idempotent support effects and ambiguous-send reconciliation.

Acceptance: Reply without updates; forget/opt-out blocks sends; no duplicate effects; failure alerts; outside-window policy reviewed.

Start: gateway; human_reply; outbox; delivery UI.

F · Menu/language/expiry

Copyable brief: Interactive nine-topic menu, language/human/emergency/consent/delete. Short-lived preferences passed to API. Consistent expired-session recovery for human/updates/chat.

Acceptance: Correct topic/language; AI initially off; command recovery; sender/session/outbox deletion; accurate Meta disclosure.

Start: gateway process/webhook; MessageInput; localisation/tests.

One worker until shared locks/storage. Use mocked errors and approved test phones. Do not promise exactly-once delivery without proof.

Prompts: referrals and inclusion.

G · Verified regional referrals

Copyable brief: Add coverage/contact channels/hours/languages/trust tiers/evidence/review date. Optional coarse region in chat/directory; no precise location. Reviewer workflow and stale/unverified labels; emergency resources separate.

Acceptance: Region/language matches; no falsely confirmed stale contacts; no exact-location requirement; audit provenance.

Start: DirectoryInput/get_directory/chat; both UIs/shared types.

H · Languages and accessibility

Copyable brief: Twi/Ga/Ewe/Hausa catalogue with professional safety/legal review. Explicit untranslated fallback; align WhatsApp/web. Keyboard/focus/screen reader/contrast/zoom/low bandwidth/quick exit evaluation against WCAG 2.2 AA.

Acceptance: Translated navigation/consent/errors; approved urgent templates; fallback disclosed; keyboard handoff/delete; mobile/screen-reader evidence.

Start: localization.tsx; page/styles; gateway/content.

Human messages/source documents are not professionally translated automatically. Consented follow-up needs a separate optional purpose/contact flow with encrypted storage, access controls and deletion.

Prompts: operations and reporting.

I · Availability and alerts

Copyable brief: Hours/availability, assignment/acknowledgement, urgent alerts/escalation timers. Approved channel without transcripts/identifiers; honest visitor delays. Failed-job/outage view, controlled audited retry and runbook.

Acceptance: Critical reaches on-call; unacknowledged escalates; true after-hours copy; no emergency promise; safe retry.

Start: queue/admin; gateway worker; settings.

J · Aggregate metrics/feedback

Copyable brief: Coarse topic/channel/outcome, queue waits/provider errors/referral clicks and optional short satisfaction. Approved retention/small-count suppression; no raw chats/session hashes/phones. Authorised reports/metric definitions.

Acceptance: Synthetic counts reconcile; small cohorts suppressed; privacy/retention documented; no ad trackers.

Start: support outcomes/events; feedback/report UI.

Partner agreements, professional review and trained staffing need accountable owners. Engineering implements workflows but cannot establish real-world coverage or partnerships.

Prompts: infrastructure and expansion.

K · CI and staging proof

Copyable brief: GitHub Actions npm ci/lint/types/builds/Python tests/audits. Build Python containers with synthetic production startup smoke. Hosted harness with URLs/temporary credentials; checks before promotion, redacted logs.

Acceptance: Fresh runner passes; images start; failure blocks; hosted auth/delete/handoff/reconnect/persistence; no real fixtures.

Start: scripts/packages/Dockerfiles; .github/workflows.

L · Shared storage/scale

Copyable brief: Active support/gateway PostgreSQL, migrations, outbox/broker/distributed limits. Preserve encryption/capabilities; worker locks/idempotency/trusted proxy attribution; staged cutover/rollback.

Acceptance: Shared sessions/cases across replicas; no races; expiry/delete; restore rehearsed; fair limits.

Start: support.py; gateway; migrations/config.

M · Later channels/scams

Copyable brief: Owner-approved SMS/USSD with signed callbacks, menus/privacy/consent/shared API. Separately evidence-based phone/screenshot-text checks, scam registry and explicit-consent reporting; no external messages without approved integration.

Acceptance: Provider menu/handoff/delete; SMS limits disclosed; evidence supports verdict; reporting opt-in.

Start: New adapter; scanner/UI/provider tests.

Access transfer and completion.

Invite the developer

- GitHub role, protected branches, PR review and issue backlog.
- Railway project/environment; both Vercel projects; owned domain/DNS.
- Scoped provider/Meta accounts through invitations/secret manager, not PDF keys.
- Blueprint, PDF, existing guide, reviewed content/partner/staff contacts and incident plan.
- Inventory web/admin/API/webhook URLs, roots/ports/volumes, SHAs and owners.

Definition of complete

All advertised features pass hosted acceptance; reviewed knowledge and emergency referrals cover nine pillars; trained staff cover promised hours; deletion/retention/encrypted recovery work; staff access attributable; WhatsApp/provider failures handled; security/content/safeguarding sign-off. Later scope excluded until delivered.

Source references

[README.md](#) · [PRODUCTION.md](#) · [VERCEL.md](#) · [VERIFICATION.md](#) · [blueprint_text.txt](#)

Official hosting references

[Railway Dockerfiles](#) · [Volumes](#) · [Private networking](#) · [Healthchecks](#) · [Vercel monorepos](#)

Documentation only; no application deployment was changed. Verify current inventory before production changes and retain synthetic release evidence.